Supplementary Material: Exact Wave-Function Solution of Completely Free Rectangular Plates: Four-Corner Kirchhoff Jump and Mode Screening

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This file is the complete Supplementary Material to the companion manuscript: §§S.1–S.5. Main-text §3.2 cites §S.1; §5.2–§5.4 cite §§S.2–S.3 (Gorman uses the same conversion as Bardell); the flexural MAC of §4.3 cites §S.5; footnotes that read "Job-level detail" point here at §S.4. Equation tags (16), (1) and (36) retain the source-paper numbering of Seok, Tiersten and Scarton.

S.1 Closed-contour four-corner Kirchhoff jump

S.1.1 The exact-differential term of Eq. (16)

Seok, Tiersten and Scarton's rectangular Part 1 Eq. (16) is a closed-contour weak-form identity on the plate boundary $\partial\Omega = c_C \cup c_N$ (clamped and natural portions). Written out in full,

$$\begin{aligned}
&\int_{c_C} \Big\{ u_3^{(0)}\, \delta\big(n_a t_{a3}^{(0)}\big) - u_{3,b}^{(0)}\, \delta\big(n_a t_{ab}^{(1)}\big) \Big\}\, ds \\
&\quad - \int_{c_N} \Big\{ \big(n_a t_{a3}^{(0)} + (n_a t_{ab}^{(1)} s_b)_{,s}\big)\, \delta u_3^{(0)} - n_a t_{ab}^{(1)} n_b\, \delta u_{3,n}^{(0)} \Big\}\, ds \\
&\quad + \int_{c_N} \big(n_a t_{ab}^{(1)} s_b\, \delta u_3^{(0)}\big)_{,s}\, ds = 0.
\end{aligned} \tag{S.1}$$

Only the third integral is an exact differential in the arclength s. Writing $\varphi := n_a t_{ab}^{(1)} s_b$ (the traction-moment contraction that reduces to the twisting moment M_{xy} on a smooth edge),

$$\int_{c_N} (\varphi\, \delta u_3^{(0)})_{,s}\, ds = \sum_{\text{corners}} \Big[\varphi\, \delta u_3^{(0)}\Big]_{\text{end}} - \Big[\varphi\, \delta u_3^{(0)}\Big]_{\text{start}}, \tag{S.2}$$

the sum running over every free–free corner where two arcs of c_N meet; interior-arc contributions cancel against the $\varphi_{,s}$ term already present in the second integral (the usual Kirchhoff effective-shear reduction). A completely free rectangle has $c_C = \varnothing$ and four such corners.

Place one corner at the local origin with both positive local axes pointing into the plate, so the two free edges meeting there are the local $+x$ and $+y$ half-axes. Consistent counterclockwise orientation of $\partial\Omega$ gives, on the vertical edge ($x = 0^+$, s increasing in $+y$), $\mathbf{n} = +\mathbf{e}_x$, $\mathbf{s} = +\mathbf{e}_y$, so $\varphi_V = t^{(1)}_{xy}$; on the horizontal edge ($y = 0^+$, s increasing in $-x$), $\mathbf{n} = +\mathbf{e}_y$, $\mathbf{s} = -\mathbf{e}_x$, so $\varphi_H = -t^{(1)}_{yx}$. Because $t^{(1)}_{ab}$ is symmetric, $\varphi_V - \varphi_H = 2\,t^{(1)}_{xy}$, so the corner's contribution to the first variation is $+2\,t^{(1)}_{xy}\,\delta u^{(0)}_3$. In the isotropic Kirchhoff limit $t^{(1)}_{xy}$ is the classical twisting moment M_{xy}, so the natural condition is $2M_{xy} = 0$ — Timoshenko's concentrated corner reaction $R = 2M_{xy}$. This single-corner trace and its classical-limit check were carried out and passed against that pre-registered criterion before any four-corner assembly was attempted.

The overall sign of that contribution is a convention rather than a result. Which edge of $\partial\Omega$ carries the + label is fixed by the global orientation chosen for Eq. (S.1), and reversing that choice reverses every boundary term of the stationarity statement together. Only the *relative* sign between the corner term and the surface terms it is added to is physical, and that relative sign is set by the same ±1 orientation factor applied corner-by-corner in §S.1.2. Fixing φ's sign as the project's other free–free derivations do (annular, variational and Green forms) writes the corner contribution as

$$\text{corner contribution} = -2\,t^{(1)}_{xy}\,\delta u^{(0)}_3\Big|_{\text{corner}}. \tag{S.3}$$

S.1.2 Four corners, and the discrete trial/test evaluation

In the discrete basis, the twisting-moment factor at a corner is the trial function's mixed second derivative $\partial^2 w/\partial x_1 \partial x_2$ there, and the paired test factor is the test function's value w there. Direct differentiation confirms that every corner term is $M^{\text{trial}}_{xy}(\text{corner})\cdot w^{\text{test}}(\text{corner})$, matching the already-validated clamped-free tip-corner term. Define, for a given trial branch ξ' and test branch ξ'',

$$\begin{aligned} p_{1t} &= \left.\frac{\partial\psi'_1}{\partial x_2}\right|_{x_2=+\pi/2}, \quad p_{1b} = \left.\frac{\partial\psi'_1}{\partial x_2}\right|_{x_2=-\pi/2}, \\ q_{0t} &= \psi''_0\big|_{x_2=+\pi/2}, \qquad q_{0b} = \psi''_0\big|_{x_2=-\pi/2}, \end{aligned} \tag{S.4}$$

and, with U_{t1}, V_{t0} the $x_1 = +L$ (tip) trial/test factors and U_{w1}, V_{w0} the $x_1 = -L$ (wall) factors, the four raw corner terms

$$\begin{aligned} \mathrm{TT} &= p_{1t}U_{t1}q_{0t}V_{t0}, \qquad \mathrm{TB} = p_{1b}U_{t1}q_{0b}V_{t0}, \\ \mathrm{WT} &= p_{1t}U_{w1}q_{0t}V_{w0}, \quad \mathrm{WB} = p_{1b}U_{w1}q_{0b}V_{w0}. \end{aligned} \tag{S.5}$$

TT and WB share the local orientation used above (outward normal along the trial axis, +2 sense) and are themselves a diagonal pair; TB and WT sit at the other diagonal, where the same closed-contour threading picks up the opposite sense, because the inward-axis Jacobian there is -1 — the discrete restatement of the classical alternating-sign corner-reaction pattern at the four corners of a free rectangular plate.

S.1.3 The parity identity, in both symmetry classes

The half-model SYM/ANTI mirror conditions fix the phase of the $m = 0$ and $m = 1$ x_2-profiles at $x_2 = \pm\pi/2$. Direct evaluation of that phase algebra gives:

- **SYM** ($m = 1$, extra $\pi/2$ phase flips sign under $x_2 \to -x_2$; $m = 0$ does not): $p_{1b} = -p_{1t}$ exactly, $q_{0b} = q_{0t}$ exactly.
- **ANTI** (roles of the two profiles swap): $p_{1b} = p_{1t}$ exactly, $q_{0b} = -q_{0t}$ exactly.

Both were confirmed numerically to working precision (the bracket $p_{1t}q_{0t} + p_{1b}q_{0b}$ reproduces 0 to $\sim 10^{-27}$ at 25 decimal digits, independent of branch, Λ, ℓ/b or ν). **The same two relations force** TB $= -$TT **and** WB $= -$WT **in both classes**, term by term rather than only in the combined bracket: in SYM, TB $= p_{1b}U_{t1}q_{0b}V_{t0} = (-p_{1t})U_{t1}(q_{0t})V_{t0} = -$TT, and identically WB $= -$WT; in ANTI, TB $= p_{1t}U_{t1}(-q_{0t})V_{t0} = -$TT and again WB $= -$WT.

Consequence 1 — why the original four-term same-sign sum is identically zero. The first free-free corner term ever assembled was the naive TT + TB + WT + WB. Substituting TB $= -$TT, WB $= -$WT gives TT $-$ TT $+$ WT $-$ WT $\equiv 0$ in both classes — confirmed numerically to 1.6×10^{-24} (SYM) and 2.8×10^{-23} (ANTI) on real assembled matrices, i.e. exactly zero to precision, not merely small. That combination contributed nothing to $\mathbf{K}$ regardless of Λ, ℓ/b, or branch.

Consequence 2 — the surviving checkerboard is exactly twice the single-point shorthand. The deployed corner term flips the sign of the diagonal pair instead of summing everything the same way:

$$\mathrm{corner}_{ff} = (T - \nu)(-2)\,(\mathrm{TT} - \mathrm{TB} - \mathrm{WT} + \mathrm{WB}). \tag{S.6}$$

Substituting the same two relations: in SYM, TT$-$TB$-$WT$+$WB $=$ TT$-(-$TT$)-$WT$+(-$WT$) =$ 2(TT $-$ WT); in ANTI the identical substitution gives the identical result, 2(TT $-$ WT). So the checkerboard bracket equals **twice** the earlier single-point candidate TT$-$WT — in both symmetry classes, by the same two-line substitution.

S.1.4 Numerical checks

- **Regression on the unmodified path.** The clamped-free branch of the out-of-plane assembler is untouched by this term (it uses its own corner, not the free-free corner). A diagnostic swap of the free-free formula into the clamped-free path reproduces the parent byte-for-byte ($\max|\mathbf{K} - \mathbf{K}_{\text{parent}}| = 0$ at the validated Seok Part 1 Table 3 SYM benchmark, $\ell/b = 1.5$, $\Lambda = 0.1551$). The solver version identifier is unchanged; only the free-free corner formula moved.

- **Algebraic gate.** Before any FE comparison was trusted, a pre-registered three-way comparison required: the naive sum genuinely $\equiv 0$; the single-point and checkerboard candidates genuinely nonzero; and $\mathbf{K}_{\text{checkerboard}} - \mathbf{K}_{\text{naive}} = 2\,(\mathbf{K}_{\text{single}} - \mathbf{K}_{\text{naive}})$ to $\sim 10^{-10}$ relative. All three passed.

- **Independent finite-element cross-check.** At the production basis, the checkerboard's nearest-to-FE dip beat both the naive-sum term (which contributes nothing) and the single-point candidate at every finite-element anchor available at the time (four points across $\ell/b = 1.5, 2.5$, SYM and ANTI). Main-text §4's five-aspect-ratio table is a later, larger-sample confirmation of the same formula.

S.1.5 What this section does not claim

The single-corner Eq. (16) trace is a first-principles derivation, checked against the classical $R = 2M_{xy}$ limit. The extension from one corner to the specific four-corner checkerboard sign pattern is argued from the closed-contour orientation logic plus the classical alternating-corner-reaction pattern, and is *validated* rather than independently re-derived corner-by-corner from Eq. (16) a second time — the same status the annular companion's corner term has relative to its own Green-identity cross-check. The parity identity is a fully first-principles algebraic fact about this project's specific ($m = 0, m = 1$) trial-function evaluations, exact in both classes. Nothing here is a claim about the in-plane formulation, which has no Kirchhoff jump term at all (Part 2 Eq. (1)/(36); see main-text §5).

S.2 In-plane nondimensionalization and Bardell mapping

S.2.1 This project's $\bar{\Omega}$

Seok, Tiersten and Scarton Part 2 Eqs. (22)–(23): $\bar{\omega} = (\pi/(2b))\sqrt{c_{66}/\rho}$, $c_{66} = E/(2(1+\nu))$, $\bar{\Omega} = \omega/\bar{\omega}$. For the FE decks ($b = 1\,\mathrm{m}$, $E = 210\,\mathrm{GPa}$, $\nu = 0.30$, $\rho = 7800\,\mathrm{kg\,m^{-3}}$) this is $\bar{\Omega} = f[\mathrm{Hz}]/804.481$, independent of ℓ/b. Half-model PLANE183, mesh-conv max 0.004%.

S.2.2 Bardell, Langley and Dunsdon (1996)

N. S. Bardell, R. S. Langley and J. M. Dunsdon, *J. Sound Vib.* **191**(3) (1996) 459–467 [3]. Their parameter is $\Omega_B = \omega a/C_L$ with $C_L^2 = E/[\rho(1-\nu^2)]$, frequencies printed under Figure 1 (F–F–F–F) rather than tabulated. Poisson ratio is not stated numerically. Their own Table 1, an exact S–S–S–S convergence study, gives the square pair $(0,1) = (1,0) = 1.859$, which equals $\pi\sqrt{(1-\nu)/2} = 1.8586$ at $\nu = 0.30$.

Plates identify as $a = 2\ell$, so $a/b = \ell/b$ and

$$\Omega_B = \bar{\Omega} \cdot \pi \cdot (\ell/b) \cdot \sqrt{(1-\nu)/2}. \tag{S.7}$$

Figure 1 caption: "(a) $a/b = 1$; (b) $a/b = 2$". Panel (a) is drawn 2:1 with unique frequencies $\{1.954, 2.961, 3.267, 4.726, 4.784, 5.205\}$; panel (b) is square with the repeated pair $2.472, 2.472$. Body text (p. 462) assigns repeats to $a/b = 1$. Mapping follows the rendered proportions, not the caption letters. The source's own Table 1 does not have this swap.

Printed F–F–F–F first six, assigned that way:

- $a/b = 1$ (square, Fig. 1(b)): 2.321, 2.472, 2.472, 2.628, 2.987, 3.452.
- $a/b = 2$ (2:1, Fig. 1(a)): 1.954, 2.961, 3.267, 4.726, 4.784, 5.205.

All twelve convert onto matched $\bar{\Omega}^*$ in main-text Table 4 (solver max miss 1.69%; independent FE max miss 0.019%).

S.3 Full in-plane FE match list

Window $\bar{\Omega} \in [0.02, 2.50]$, five aspect ratios, both parities. Count $5+4+8+6+10+7+13+10+16+13 = 92$ is every non-rigid FE target in that window. Table S.1 lists the unique MATCH after collapsing double-dips onto the same FE partner. Production discovery used $n_{\mathrm{cpair}} = 3$ and a 3% cut; sixteen remaining targets were recovered at $n_{\mathrm{cpair}} = 5$ in a ± 0.05 window centred on the FE value.

Table S.1: Unique in-plane MATCH against independent PLANE183 half-model FE, $\bar{\Omega} \in [0.02, 2.50]$, all five aspect ratios and both parities. Job numbers in the last column: 2456002 is $n_{\text{cpair}} = 3$ production discovery; 2456013 is the $n_{\text{cpair}} = 5$ mop-up of the sixteen remaining targets.

ℓ/b	class	$\bar{\Omega}^*$	$\bar{\Omega}_{\text{FE}}$	miss	job
1.0	SYM	1.3300	1.32984	0.01%	2456002
1.0	SYM	1.4100	1.41422	0.30%	2456002
1.0	SYM	1.6100	1.60733	0.17%	2456002
1.0	SYM	1.8600	1.85745	0.14%	2456002
1.0	SYM	2.0000	2.00319	0.16%	2456002
1.0	ANTI	1.2400	1.24858	0.69%	2456002
1.0	ANTI	1.3298	1.32984	0.00%	2456013
1.0	ANTI	2.0000	2.00319	0.16%	2456002
1.0	ANTI	2.3200	2.31523	0.21%	2456002
1.5	SYM	1.0500	1.04551	0.43%	2456002
1.5	SYM	1.4100	1.41236	0.17%	2456002
1.5	SYM	1.4243	1.42432	0.00%	2456013
1.5	SYM	1.6300	1.62439	0.35%	2456002
1.5	SYM	1.6900	1.69877	0.52%	2456002
1.5	SYM	2.1100	2.10886	0.05%	2456002
1.5	SYM	2.2200	2.22185	0.08%	2456002
1.5	SYM	2.2800	2.28542	0.24%	2456002
1.5	ANTI	0.7800	0.78782	0.99%	2456002
1.5	ANTI	1.0333	1.03335	0.00%	2456013
1.5	ANTI	1.5712	1.57124	0.00%	2456013
1.5	ANTI	1.6400	1.64765	0.46%	2456002
1.5	ANTI	2.2025	2.20253	0.00%	2456013
1.5	ANTI	2.2400	2.24065	0.03%	2456002
2.0	SYM	0.8100	0.79652	1.69%	2456002
2.0	SYM	1.4000	1.40011	0.01%	2456002
2.0	SYM	1.4142	1.41422	0.00%	2456013
2.0	SYM	1.4400	1.44333	0.23%	2456002
2.0	SYM	1.6500	1.65355	0.21%	2456002
2.0	SYM	1.7500	1.77390	1.35%	2456002
2.0	SYM	1.8100	1.81566	0.31%	2456002
2.0	SYM	2.0000	2.00440	0.22%	2456002
2.0	SYM	2.3000	2.29856	0.06%	2456002
2.0	SYM	2.3300	2.33104	0.04%	2456002
2.0	ANTI	0.5200	0.52557	1.06%	2456002
2.0	ANTI	0.8800	0.87891	0.12%	2456002
2.0	ANTI	1.2715	1.27148	0.00%	2456013
2.0	ANTI	1.2800	1.28703	0.55%	2456002
2.0	ANTI	1.7400	1.73452	0.32%	2456002
2.0	ANTI	1.8400	1.84467	0.25%	2456002
2.0	ANTI	2.1400	2.13652	0.16%	2456002
2.5	SYM	0.6400	0.64049	0.08%	2456002
2.5	SYM	1.2200	1.22293	0.24%	2456002
2.5	SYM	1.4100	1.41225	0.16%	2456002
2.5	SYM	1.4266	1.42658	0.00%	2456013
2.5	SYM	1.5400	1.53449	0.36%	2456002
2.5	SYM	1.6700	1.67003	0.00%	2456002
2.5	SYM	1.6900	1.69350	0.21%	2456002
2.5	SYM	1.7800	1.78140	0.08%	2456002
2.5	SYM	2.0500	2.04232	0.38%	2456002

Table S.1: Unique in-plane MATCH (continued).

ℓ/b	class	$\bar{\Omega}^*$	$\bar{\Omega}_{\mathrm{FE}}$	miss	job
2.5	SYM	2.0700	2.07671	0.32%	2456002
2.5	SYM	2.1300	2.13048	0.02%	2456002
2.5	SYM	2.3900	2.38996	0.00%	2456002
2.5	SYM	2.4047	2.40468	0.00%	2456013
2.5	ANTI	0.3800	0.37585	1.10%	2456002
2.5	ANTI	0.7200	0.71776	0.31%	2456002
2.5	ANTI	1.0600	1.06883	0.83%	2456002
2.5	ANTI	1.1125	1.11248	0.00%	2456013
2.5	ANTI	1.4400	1.44246	0.17%	2456002
2.5	ANTI	1.5400	1.53132	0.57%	2456002
2.5	ANTI	1.8000	1.80308	0.17%	2456002
2.5	ANTI	2.0000	1.99255	0.37%	2456002
2.5	ANTI	2.2000	2.14745	2.45%	2456002
2.5	ANTI	2.2800	2.34878	2.93%	2456002
3.0	SYM	0.5400	0.53502	0.93%	2456002
3.0	SYM	1.0600	1.04554	1.38%	2456002
3.0	SYM	1.4142	1.41422	0.00%	2456013
3.0	SYM	1.4199	1.41791	0.14%	2456013
3.0	SYM	1.4200	1.41978	0.02%	2456002
3.0	SYM	1.6000	1.60066	0.04%	2456002
3.0	SYM	1.6635	1.66346	0.00%	2456013
3.0	SYM	1.6900	1.67627	0.82%	2456002
3.0	SYM	1.8000	1.80401	0.22%	2456002
3.0	SYM	1.8400	1.83870	0.07%	2456002
3.0	SYM	2.0000	2.00478	0.24%	2456002
3.0	SYM	2.1600	2.16287	0.13%	2456002
3.0	SYM	2.2100	2.21145	0.07%	2456002
3.0	SYM	2.2700	2.27554	0.24%	2456002
3.0	SYM	2.4300	2.44758	0.72%	2456002
3.0	SYM	2.4600	2.46034	0.01%	2456002
3.0	ANTI	0.2800	0.28169	0.60%	2456002
3.0	ANTI	0.5800	0.57789	0.37%	2456002
3.0	ANTI	0.8800	0.88739	0.83%	2456002
3.0	ANTI	1.0315	1.03153	0.00%	2456013
3.0	ANTI	1.2702	1.27024	0.00%	2456013
3.0	ANTI	1.3000	1.29465	0.41%	2456002
3.0	ANTI	1.5800	1.57876	0.08%	2456002
3.0	ANTI	1.6600	1.66509	0.31%	2456002
3.0	ANTI	1.9000	1.91234	0.65%	2456002
3.0	ANTI	2.0000	1.99856	0.07%	2456002
3.0	ANTI	2.2400	2.25049	0.47%	2456002
3.0	ANTI	2.2800	2.27620	0.17%	2456002
3.0	ANTI	2.4937	2.49371	0.00%	2456013

S.4 Cluster job-number provenance

The main text states every result in prose, with no job number in its own running text; this section is the audit trail. Each entry is keyed to the main-text location that cites this section.

- **§3.2 / §S.1, checkerboard corner.** Deployed in the out-of-plane assembler under the free-free option only. Algebraic gate and four-point FE cross-check: `probe_rect_ff_oop_derived_vs_fix_2026-09-03.py`. Clamped-free regression at $\ell/b = 1.5$, $\Lambda = 0.1551$: $\max|\mathbf{K} - \mathbf{K}_{\text{parent}}| = 0$. `SOLVER_VERSION 2026-07-10.s10`, unbumped.

- **§3.3, Screen B freeze.** Persist $n_{\text{cpair}} = 3$, $|\Delta\Lambda| \leq 0.01$, frozen before the extra-ℓ/b geometries were run. Not retuned against FE. Post-hoc continuum sensitivity check: job 2457815 rescored all 148 published flexural and in-plane matches against the frozen fixed-step scan (all pass); job 2458006 re-scanned the three that sat at the scan's outer edge with a 4× finer grid; job 2458121 settled them with a golden-section continuum minimisation. Two main-text Table 2 entries ($\ell/b = 1.5$ SYM 5.380, $\ell/b = 2.5$ ANTI 6.260) exceed the frozen cut at the continuum optimum ($|\Delta\Lambda| = 0.0103, 0.0110$); one in-plane entry ($\ell/b = 3.0$ ANTI $\bar{\Omega}^* = 2.2400$, Table S.1) does not ($|\Delta\bar{\Omega}| = 0.0094$).

- **§4, flexural FE.** $\ell/b = 1.5$ half-model, job 2398617 lineage; $\ell/b = 1.0, 2.5$, P5 Phase A queue; $\ell/b = 2.0, 3.0$, job 2454654. Primary refine 1.0/1.5/2.5: job 2453874 (12 unique MATCH at 3%); 2.0/3.0: job 2454701 (12 unique MATCH at 3%). Persist-only SYM hits at 0.543 and 0.670: job 2454702, both HIT at 0.00%. High-Λ 1.0/1.5/2.5: job 2453875; 2.0/3.0: job 2454703 (machine zeros tagged `ZERO_ART`).

- **§4.4,** $\Lambda_{\text{FE}} = 1.982$ **unmatched curiosity.** Job 2455675 reopened $\ell/b = 1.0$ SYM 1.98249 at persist basis $(6, 3)$, step 0.0002 in $[1.96249, 2.00249]$: global min $\sigma = 1.702 \times 10^{-6}$ at the window's right edge, not flanked (pre-registered reopen test failed; miss 1.009%). Job 2456003 widened the scan to $[1.73, 2.23]$ (step 0.002): σ stays within one order of magnitude of its floor (ratio 3.12), no dip within 3% of 1.98249. Flat-plateau confirmed; Screen B not retuned.

- **§4.5, Leissa 1973.** Conversion of Table C15 $\lambda = \omega a^2\sqrt{\rho h/D}$ through the same $K = 24.6644\,\text{Hz}$ as §4; no cluster job. Fifteen overlapping entries, each paired with its own parity under Leissa's (m, n) rather than with the nearest frequency; three named near-misses (square $\lambda = 19.789$; $\ell/b = 1.5$ $\lambda = 58.201$ and $\lambda = 67.494$) discussed in the main text.

- **§5.1, in-plane no-corner.** Seok Part 2 Eq. (1)/(36); assembler is wall+tip only. Free-free option is an edge-list swap. Default clamped-free bit-identical (sandbox $\max|\mathbf{K}_{\text{default}} - \mathbf{K}_{\text{cf}}| = 0$).

- **§5.2, in-plane FE.** Job 2455569, PLANE183 half-model at five aspect ratios, 5/5 OK, mesh-conv max 0.004%, $\bar{\Omega} = f/804.481$.

- **§5.2, in-plane discovery.** Job 2455570 ($n_{\text{cpair}} = 0$, 5% cut): 38 unique MATCH, ANTI branch weak. Jobs 2455676/77/78 traced this to basis starvation, not missing physics. Job 2456002 (production re-discovery at $n_{\text{cpair}} = 3$, 3% cut): 76 unique MATCH. Job 2456013 (targeted mop-up, $n_{\text{cpair}} = 5$, ± 0.05 on the remaining 16 FE values): 16/16 recovered, 0.00–0.14% miss. Net 92/92 in $\bar{\Omega} \in [0.02, 2.50]$. Two apparent losses between the 38- and 76-match passes, $(2.0, \text{ANTI}, 1.27148)$ and $(3.0, \text{SYM}, 1.66346)$, are both present in the 92 (root-crowding). Job 2455677's three ANTI "NO DIP" results ($\ell/b = 1.0$ ANTI) are the same seeding artifact: their true FE targets 1.24858/2.00319/2.31523 are clean matches in the 92.

- **§5.3, Bardell 1996.** Frequencies read from Figure 1 of the source PDF (mode-plot labels, not a table). Aspect-ratio mapping by plate proportions and repeated-frequency physics, not the caption letters. Conversion Eq. (S.7) at $\nu = 0.30$, confirmed by the source's Table 1 S–S–S–S exact 1.859. Matched $\bar{\Omega}^*$ from jobs 2456002 and 2456013; FE from job 2455569. 12/12 published F–F–F–F first-six values matched.

- **§5.4, Gorman 2004.** Tabulated λ^2 doubled before Eq. (S.7) (Gorman's a is the half-length); nineteen in-window eigenvalues, no cluster job.
- **§4.3, flexural MAC.** Eigenvectors from job 2456738 (five SHELL281 MAC-extract decks, 5/5 `QUEUE_OK`). Convert $f_{\mathrm{Hz}}/24.6644$ in Python; the deck-printed Λ column is Instance 3 of the project's `*VWRITE`/KFAC bug and is not used. Off-cluster reconstruction: `probe_rect_ff_oop_mac_2026-09-05.py` and `probe_rect_ff_oop_mac_strengthen_2026-09-05.py`. Bar 0.744 is the annular companion's production call, unre-tuned. Job 2456779 (dps= 40 persist golden section on the seven main-text Table 1 ANTI rows that do not meet the bar): interior minima within 8×10^{-4} of the published Λ^*; persist MAC against the frequency partner does not cross 0.744 (gate A of that probe; the bar is not retuned). `SOLVER_VERSION 2026-07-10.s10`, unbumped.

S.5 Flexural MAC against half-model eigenvectors

The identity-weighted MAC is $\mathrm{MAC} = |\langle u_Z, w\rangle|^2/(\|u_Z\|^2\|w\|^2)$ on the phase-aligned real part of the reconstructed analytical w, evaluated at the FE nodes of the half-model ($X \in [-\ell, \ell]$, $Y \in [0, b]$, $x_1 = X\pi/(2b)$, $x_2 = Y\pi/(2b)$). The bar 0.744 is the annular companion's production call [4], not retuned against these data. Table S.2 lists every main-text Tables 1–2 pair that meets the bar against its frequency-matched same-parity FE mode: 36 of the 56 published rows, 25 of 29 symmetric and 11 of 27 antisymmetric. Included are three persist-basis recoveries, six $n_{\mathrm{cpair}} = 6$ recoveries (three of those after discarding a machine-zero singular vector at $\sigma \sim 10^{-14}$, the known $\Lambda = 4, 6$ artifact), and the persist-only $\ell/b = 3.0$ SYM 0.6698 row whose persist kernel is two-dimensional (the next singular vector is the physical shape). Known Screen B leaks are not in the table: SYM 0.41 has MAC ≤ 0.05 against every FE mode of that parity; ANTI 0.46 stays below the bar except at $\ell/b = 3.0$, where it coincides with the real first ANTI and scores 1.000 as that mode. Rows that remain below the bar — twenty rows, including the $\ell/b = 2.5$ first ANTI (MAC saturates at 0.655), the $\ell/b = 1.5$ ANTI 2.124 weakest primary, and both ‡-marked rows of main-text Table 2 — are frequency matches only. The lowest MAC in the table is 0.751 and the highest value reached by any excluded row is 0.655, so the confirmed set is unchanged for any bar between those two.

Table S.2: MAC of analytical w against the frequency-matched SHELL281 U_Z. Production basis unless noted. Bar 0.744, not retuned.

ℓ/b	class	Λ^*	Λ_{FE}	MAC	basis	note
1.0	ANTI	1.366	1.35288	0.941	prod.	1st ANTI; Leissa $(2, 2)$
1.5	SYM	0.964	0.96347	1.000	prod.	1st SYM; Leissa $(3, 1)$
1.5	SYM	2.254	2.24373	1.000	prod.	Leissa $(1, 3)$
1.5	ANTI	0.906	0.89838	0.852	prod.	1st ANTI; Leissa $(2, 2)$
2.0	SYM	0.5434	0.54338	1.000	persist	persist-only frequency
2.0	SYM	1.510	1.50735	0.849	persist	
2.0	SYM	2.234	2.22558	0.999	prod.	
2.0	ANTI	0.674	0.66858	0.751	prod.	1st ANTI
2.5	SYM	0.348	0.34766	0.999	prod.	1st SYM; Leissa $(3, 1)$
2.5	SYM	0.978	0.96559	0.871	persist	

ℓ/b	class	Λ^*	Λ_{FE}	MAC	basis	note
2.5	SYM	1.888	1.88370	0.999	prod.	
2.5	SYM	2.278	2.27059	0.999	prod.	
2.5	SYM	3.190	3.19522	0.915	$n_{\mathrm{cpair}} = 6$	close pair; see text
2.5	SYM	4.180	4.13819	0.956	$n_{\mathrm{cpair}} = 6$	skip machine-zero SV
2.5	ANTI	1.918	1.90392	1.000	prod.	Leissa $(4, 2)$
2.5	ANTI	6.400	6.36131	1.000	prod.	
3.0	SYM	0.242	0.24127	0.959	prod.	1st SYM
3.0	SYM	0.6698	0.66984	1.000	persist	next SV of a 2-D kernel
3.0	SYM	1.320	1.31785	1.000	prod.	
3.0	SYM	2.162	2.15417	0.779	persist	
3.0	SYM	2.256	2.24841	0.999	prod.	
3.0	SYM	2.480	2.45935	0.993	$n_{\mathrm{cpair}} = 6$	
3.0	SYM	3.340	3.32528	0.993	prod.	
3.0	SYM	3.660	3.62467	0.964	$n_{\mathrm{cpair}} = 6$	skip machine-zero SV
3.0	SYM	5.660	5.59746	0.926	$n_{\mathrm{cpair}} = 6$	skip machine-zero SV
3.0	ANTI	0.444	0.44044	1.000	prod.	1st ANTI; 0.46-family
3.0	ANTI	3.160	3.14253	0.999	prod.	
3.0	ANTI	5.560	5.51872	0.963	prod.	
3.0	ANTI	6.380	6.34339	1.000	prod.	
1.0	SYM	2.460	2.45439	0.997	prod.	Leissa $(3, 1)$
1.0	SYM	6.190	6.16026	0.867	$n_{\mathrm{cpair}} = 6$	
1.0	SYM	6.460	6.36553	1.000	prod.	
1.5	ANTI	3.860	3.83354	0.993	prod.	
2.0	SYM	3.000	2.99812	0.862	prod.	
2.0	SYM	3.650	3.62227	1.000	prod.	
2.0	ANTI	2.580	2.55180	0.999	prod.	

References

[1] J. Seok, H.F. Tiersten, H.A. Scarton, "Free vibrations of rectangular cantilever plates. Part 1: out-of-plane motion," J. Sound Vib. 271 (2004) 131–146. `https://doi.org/10.1016/S0022-460X(03)00365-1`.

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[3] N.S. Bardell, R.S. Langley, J.M. Dunsdon, "On the free in-plane vibration of isotropic rectangular plates," J. Sound Vib. 191 (3) (1996) 459–467. `https://doi.org/10.1006/jsvi.1996.0134`.

[4] G. McCune, D. Stutts, "Exact wave-function solution and spurious-mode discrimination for completely free annular sector plates," companion paper, submitted simultaneously to this journal (2026).